

Question #1 of 65

Question ID: 1473617

The contract price of a forward contract is:

- A) the price that makes the contract a zero-value investment at initiation. 
- B) always the present value of the expected future spot price. 
- C) determined at the settlement date. 

Explanation

The contract price can be an interest rate, discount, yield to maturity, or exchange rate. The forward price is the future value of the spot price adjusted for any periodic payments expected from the asset. An example of when the forward price may be less than the spot price is in the case of an equity index contract where the dividend yield is greater than the risk-free rate.

(Module 30.1, LOS 30.b)

Question #2 of 65

Question ID: 1473635

Consider a 9-month forward contract on a 10-year 7% Treasury note just issued at par. The effective annual risk-free rate is 5% over the near term and the first coupon is to be paid in 182 days. The price of the forward is *closest* to:

- A) 965.84. 
- B) 1,037.27. 
- C) 1,001.84. 

Explanation

The forward price is calculated as the bond price minus the present value of the coupon, times one plus the risk-free rate for the term of the forward.

$$(1,000 - 35/1.05^{182/365}) 1.05^{9/12} = \$1,001.84$$

(Module 30.3, LOS 30.d)

Question #3 of 65

Question ID: 1473670

In which type of swap contract is notional principal *most likely* to be exchanged at initiation?

- A) Interest rate swap. 
- B) Currency swap. 
- C) Equity swap. 

Explanation

Notional principal is typically exchanged at initiation of a currency swap, but not in an interest rate swap or equity swap.

(Module 30.7, LOS 30.f)

Question #4 of 65

Question ID: 1473660

Which of the following is *equivalent* to a plain vanilla receive-fixed interest rate swap?

- A) A long position in a bond coupled with the issuance of a floating rate note. 
- B) A short position in a bond coupled with a long position in a floating rate note. 
- C) A short position in a bond coupled with the issuance of a floating rate note. 

Explanation

A long position in a fixed rate bond receives fixed coupons. The short floating rate note requires floating-rate payments. Together, these are the same cash flow as a receive-fixed swap.

(Module 30.6, LOS 30.e)

Question #5 of 65

Question ID: 1473653

The price of an interest rate swap is the:

- A) market value of the swap. 
- B) fixed rate of interest. 
- C) cost to purchase a swap. 

Explanation

The price of an interest rate swap is quoted as the rate on the fixed-rate payments. The floating rate is a known reference rate but does not need to be quoted.

(Module 30.6, LOS 30.e)

Question #6 of 65

Question ID: 1473666

A \$10 million 2-year semi-annual-pay MRR-based interest-rate swap was initiated 180 days ago when swap fixed rate was 3.8%. The fixed rate on the swap is now 3.4% and the term structure is as follows:

Days	MRR	Discount Factor
180	3.00%	0.98522
360	3.20%	0.96899
540	3.40%	0.95148
720	4.00%	0.92593

Value of the swap to the payer is *closest* to:

A) -\$29,229.



B) -\$45,633.



C) -\$58,114.

**Explanation**

Sum of the discount factors for the three settlement dates remaining (180,360 and 540 days away) = $0.98522 + 0.96899 + 0.95148 = 2.9057$

Value to payer = $2.9057 \times [(0.034 - 0.038)/2] \times \$10 \text{ million} = -\$58,114$

(Module 30.6, LOS 30.e)

Question #7 of 65

Question ID: 1473643

A company has chosen to use a 6 x 9 FRA expiring in 6 months to mitigate the risk of paying a floating coupon on the bond issue. The current term structure for MRR is as follows:

Term	Interest Rate
180 days	5.65%
270 days	5.95%

What is the price of this forward rate agreement (FRA)?

A) 3.19%



B) 6.37%



C) \$6.37



Explanation

The price of an FRA is the fixed rate. To determine the FRA's fixed rate, the following formula should be used:

$$\begin{aligned}\text{FRA price} &= \left(\frac{P_{Y_{1+r}}}{P_{X_{1+r}}} - 1 \right) \left(\frac{360}{Y-X} \right) \\ &= \left[\frac{1 + .0595 \left(\frac{270}{360} \right)}{1 + .0565 \left(\frac{180}{360} \right)} - 1 \right] \left(\frac{360}{90} \right) = \underline{\underline{0.0637}}\end{aligned}$$

The FRA's fixed rate would be quoted as 6.37%.

The price of an FRA is given as a rate percentage, never as a dollar amount.

(Module 30.4, LOS 30.c)

Question #8 of 65

Question ID: 1473619

The price of a forward contract:

A) must be equal to the market price at contract termination.



B) is equal to the value of the contract in equilibrium.



C) is the settlement price for the underlying asset.



Explanation

The price of a forward contract is the price of the underlying asset that the long will pay to the short at settlement (for a deliverable contract). The value of a forward contract comes from the difference between the forward contract price and the market price for the underlying asset. This difference between price and value is a key concept to understand. A forward contract has only one price, which applies to both the long and to the short.

(Module 30.1, LOS 30.b)

Question #9 of 65

Question ID: 1473628

Over the life of a swap, the price of the swap:

A) fluctuates with changes in the yield curve.



B) is approximately equal to the market value of the swap.



C) does not change.



Explanation

The price of a swap, quoted as the fixed rate in the swap, is determined at contract initiation and remains fixed for the life of the swap.

(Module 30.1, LOS 30.b)

Question #10 of 65

Question ID: 1473641

Calculate the price (expressed as an annualized rate) of a 1x4 forward rate agreement (FRA) if the current 30-day rate is 5% and the 120-day rate is 7%.

A) 6.86%.



B) 7.63%.



C) 7.47%.



Explanation

A 1x4 FRA is a 90-day loan, 30 days from today.

The actual rate on the 30-day loan is: $R_{30} = 0.05 \times 30/360 = 0.004167$

The actual rate on the 120-day loan is: $R_{120} = 0.07 \times 120/360 = 0.02333$

$FR(30,90) = [(1 + R_{120}) / (1 + R_{30})] - 1 = (1.023333 / 1.004167) - 1 = 0.0190871$

The annualized 90-day rate = $0.0190871 \times 360/90 = .07634 = 7.63\%$

(Module 30.4, LOS 30.c)

Question #11 of 65

Question ID: 1473634

Calculate the price of a 200-day forward contract on an 8%, semi-annual, U.S. Treasury bond with a spot price of \$1,310. Next coupon payment will be made in 150 days. The annual risk-free rate is 5%.

A) \$1,270.79.



B) \$1,333.50.



C) \$1,305.22.



Explanation

Coupon = $(1,000 \times 0.08) / 2 = \40.00

Present value of coupon payment = $\$40.00 / 1.05^{150/365} = \39.21

Forward price on the fixed income security = $(\$1,310 - \$39.21) \times (1.05)^{200/365} = \$1,305.22$

(Module 30.3, LOS 30.d)

Question #12 of 65

Question ID: 1473625

The no-arbitrage price of a futures contract with a spot rate of 990, a time to maturity of 2 years, and a risk-free-rate of 5% is *closest* to:

A) 1040



B) 792



C) 1091



Explanation

The no-arbitrage price of a futures contract is based on the spot rate, the time to maturity, and the risk-free-rate.

FP	$= S_0 \times (1 + R_f)^T$
	$= 990(1.05)^2$
	$= 1091$

(Module 30.1, LOS 30.b)

Question #13 of 65

Question ID: 1473659

The fixed-rate receiver in a plain vanilla interest rate swap has a position equivalent to a series of:

A) long interest-rate puts and short interest-rate calls.



B) short interest-puts and long interest-rate calls.



C) long interest-rate puts.



Explanation

The fixed-rate receiver has profits when short rates fall and losses when short rates rise, equivalent to buying puts and writing calls.

(Module 30.6, LOS 30.e)

Question #14 of 65

Question ID: 1473649

What is the value of a 6.00% 1x4 (30 days x 120 days) forward rate agreement (FRA) with a principal amount of \$2,000,000, 10 days after initiation if the MRR curve at that time reflects the following:

110-day MRR = 6.15% and

20-day MRR = 6.05%?

A) \$700.00.



B) \$767.40.



C) \$745.76.



Explanation

The current 90-day forward rate at the settlement date, 20 days from now is:

$$([1 + (0.0615 \times 110/360)]/[1 + (0.0605 \times 20/360)] - 1) \times 360/90 = 0.061517$$

The interest difference on a \$2 million, 90-day loan made 20 days from now at the above rate compared to the FRA rate of 6.0% is:

$$[(0.061517 \times 90/360) - (0.060 \times 90/360)] \times 2,000,000 = \$758.50$$

Discount this amount at the current 110-day rate:

$$758.50/[1 + (0.0615 \times 110/360)] = \$745.76$$

(Module 30.5, LOS 30.c)

Abel Smith works in the Treasury Department of OTS Ltd. OTS is an international construction firm, based in the United States. OTS hopes to raise €100 million through the issuance of a €100 million one-year fixed rate bond but is concerned about the currency risk exposure.

TNA Bank proposed a one year EUR-USD currency swap with semi-annual settlements to OTS to mitigate the exchange rate risk. The notional principal would be €100 million. The bank provides the following information:

Exhibit 1: MRR Spot Rate (annualized)

Days	\$ MRR	PV Factors	Days	MRR	PV Factors
180	0.70%	0.9965	180	1.50%	0.9926
360	1.00%	0.9901	360	2.00%	0.9804

Using the information in Exhibit 1, the bank calculates that the currency swap's fixed rates are 0.85% on the USD and 1.75% on the euro.

The term structure of MRR and MRR spot rates three months after the swap initiation is shown in Exhibit 2:

Exhibit 2: MRR Spot Rate (annualized)

Days	\$ MRR	PV Factors	Days	MRR	PV Factors
90	0.85%	0.9979	90	1.30%	0.9968
270	1.20%	0.9911	270	2.30%	0.9830

Exhibit 3: Exchange Rates

Time	Exchange Rate	Time	Exchange Rate
Swap initiation	€1: USD1.43	1st settlement date	€1: USD1.55
90 days after swap initiation	€1: USD1.48	2nd settlement date	€1: USD1.55

OTS has an investment portfolio with similar weighting as the S&P500. Smith believes that the U.S. equity market could suffer further declines and OTS could hedge the equity risk using an equity swap. Smith obtained the outlook of the U.S. equity market in Exhibit 4.

Exhibit 4: Performance of S&P500 for the Next Three Quarters

Time	S&P Index Level
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Swap initiation	1,190
1st quarter	1,000
2nd quarter	980
3rd quarter	1,050
4th quarter	1,130

Smith is proposing for OTS to be the party paying the equity return on a USD500 million one-year equity swap. OTS will be receiving fixed rate of 1% on a semi-annual basis.

Smith makes two comments:

Comment 1: A fixed-rate payer in an equity swap would not have to pay more than the fixed rate each period.

Comment 2: Compared to an interest rate swap, the first payment in an equity swap will not be known at initiation.

Question #15 - 18 of 65

Question ID: 1473677

Has the bank correctly calculated the fixed rates on the currency swap?

A) Both fixed rates are incorrectly calculated.



B) One of the two fixed rates is incorrectly calculated.



C) Both fixed rates are correctly calculated.



Explanation

Fixed rate for the USD:

$$\frac{1-0.9901}{0.9965+0.9901} = 0.00498 = 0.498\%$$

In annual terms, the fixed rate for the USD = $0.498\% \times \frac{360}{180} = 0.997\%$

Fixed rate for the euro:

$$\frac{1-0.9804}{0.9926+0.9804} = 0.00993 = 0.993\%$$

In annual terms, the fixed rate for the euro = $0.993\% \times \frac{360}{180} = 1.987\%$

The fixed rates quoted by the bank are both incorrectly calculated.

(Module 30.7, LOS 30.f)

Question #16 - 18 of 65

Question ID: 1473678

If OTS decides on a fixed for fixed currency swap, what is the market value of the swap to OTS three months after swap initiation?

- A) The value of the euro payments is €97.68m and converting the euro value to USD using €1: USD1.55, the swap has a positive to OTS. 
- B) The value of the euro payments is €100.27m and converting the euro value to USD using €1: USD1.48, the swap has a positive to OTS. 
- C) The value of the euro payments is €101.8m and converting the euro value to USD using €1: USD1.43, the swap has a positive to OTS. 

Explanation

To mitigate currency risk, OTS would be the party paying the fixed rate on USD and receiving the fixed rate on euro. The notional principal is €100m to USD143 million. Note this is using the \$/€ of 1.43 at swap initiation.

The value of the USD fixed payments is equivalent to the value of a USD143m fixed rate bond that pays a coupon in 90 days' time and a coupon plus face value in 270 days' time. The coupon rate would be 0.997% as computed in the solution to question 1.

The coupons on the bond = $0.00997 \times (180 / 360) \times \$143\text{m} = \$712,855$

Maturity	\$ MRR	Discount Factor	CF	DCF
90	0.85%	0.9979	\$712,855	711,358
270	1.20%	0.9911	\$143,712,855	<u>142,433,811</u>
PV \$ Fixed				\$143,145,169

The value of the Euro fixed payments is equivalent to the value of a €100m fixed rate bond that pays a coupon in 90 days' time and a coupon plus face value in 270 days' time. The coupon rate would be 1.987% as computed in the solution to question 1.

The coupons on the bond = $0.01987 \times (180 / 360) \times €100\text{m} = €993,500$

Maturity	\$ MRR	Discount Factor	CF	DCF
90	1.30%	0.9968	€993,500	990,321
270	2.30%	0.9830	€100,993,500	<u>99,276,611</u>
PV € Fixed				€100,266,932

OTS is based in the U.S. but has borrowed in €'s. OTS would therefore take the position of receiving €'s and paying \$'s in the currency swap.

Convert the € fixed coupon bond into \$'s at the date of the valuation using the exchange rate \$/€ = 1.48

$€100,266,932 \times 1.48 = \$148,395,059$

The value to the party paying \$'s and receiving €'s = $\$148,395,059 - \$143,145,169 = \$5,249,890$

Note that no actual computations were required to answer this question if you spotted that answer B was the only solution to use the correct exchange rate.

Question #17 - 18 of 65

Question ID: 1473679

Using the information in Exhibit 2, what is the market value of the equity swap to OTS three months after swap initiation?

- A) +USD80.35 million.
- B) +USD85.33 million.
- C) +USD88.76 million.

**Explanation**

OTS is paying the equity return and the value is:

$$(1000 / 1190) \times \text{USD}500\text{m} = \text{USD}420.17 \text{ million}$$

The value of fixed payments is equivalent to the value of a one-year fixed coupon bond with 0.5% semi-annual coupon. The value of bond is the present value of the two coupon payments and the par value:

$$[(0.005 \times 0.9979) + (1.005 \times 0.9911)] \times 500\text{m} = \text{USD } 500.52 \text{ million.}$$

$$\text{Value to OTS: } \text{USD}500.52\text{m} - \text{USD}420.17\text{m} = \text{USD}80.35 \text{ million}$$

(Module 30.8, LOS 30.g)

Question #18 - 18 of 65

Question ID: 1473680

How many of Smith's comments are correct?

- A) Neither comment is correct.
- B) One comment is correct.
- C) Both comments are correct.

**Explanation**

When the index declines, the fixed rate payer would pay the negative return in addition to the fixed rate and, hence, will suffer a loss greater than the fixed rate. There is no way of knowing the first payment on an equity swap because we do not know the value of the equity index on the payment date. The floating rate for the first settlement also known at time 0 (known as advance set, paid in arrears).

(Module 30.8, LOS 30.g)

Question #19 of 65

Question ID: 1473633

An index is currently 876, the risk-free rate (R_f) is 7%, and the dividend yield on the index portfolio is 1.8%. Assuming that these are continuously compounded yields, the price of an 18-month index future is *closest* to:

A) 943.0.**B)** 947.1.**C)** 945.2.**Explanation**

$$FP = 876 e^{(0.07-0.018)1.5} = 947.1.$$

(Module 30.2, LOS 30.a)

Question #20 of 65

Question ID: 1473683

Consider a 1-year, \$5 million semiannual-pay fixed-rate equity swap initiated when the equity index is 750 and swap fixed rate is 3.7%. Equity index was at 760 at first settlement. It is now 270 days since inception of the swap and the index is at 767, 90-day MRR is 3.4% (DF = 0.99157) and 270-day MRR is 3.7% (DF = 0.9730). What is the value of the swap to the fixed-rate payer?

A) -\$2,726.**B)** \$3,478.**C)** -\$3,520.**Explanation**

For \$100 notional, the value of the equity side is $(767/760) \times \$100 = \100.921

Value of the semiannual -pay, fixed rate bond with 3.7% annual coupon = $[100 + 3.7/2] \times 0.99157 = \100.9914

Value of pay-fixed side = value of equity - value of fixed rate bond = $\$100.921 - \$100.9914 = -\$0.0704$ (per \$100 notional).

For \$5 million notional, value = $50,000 \times -0.0704 = -\$3,520$

(Module 30.8, LOS 30.g)

Question #21 of 65

Question ID: 1473672

Consider a one-year currency swap with semiannual payments. The payments are in U.S. dollars and euros. The current exchange rate of the euro is \$1.30 and interest rates are

	180 days	360 days
USD MRR	5.6%	6.0%
MRR	4.8%	5.4%

What is the fixed rate in euros?

A) 2.659%.



B) 5.245%.



C) 5.318%.

**Explanation**

The present values of 1 euro received in 180 days and 1 euro received in 360 days are:

$$1/(1 + 0.048 \times (180/360)) = 0.9766 \text{ and } 1/1.054 = 0.9488$$

The fixed rate in euros is $(1 - 0.9488) / (0.9766 + 0.9488) = 0.026592 \times (360/180) = 5.318\%$.

The notional principal is $100,000/1.30 = 76,923$ euros.

(Module 30.7, LOS 30.f)

Question #22 of 65

Question ID: 1473665

If the one year spot rate is 5%, the two-year spot rate is 5.5%, and the three year spot rate is 6%, the fixed rate on a 3-year annual pay swap is *closest* to:

A) 1.99%.



B) 4.50%.



C) 5.65%.

**Explanation**

The fixed rate on the swap is:

$$\frac{1 - \frac{1}{1 + 0.06(3)}}{\frac{1}{1.05} + \frac{1}{1 + 0.055(2)} + \frac{1}{1 + 0.06(3)}}$$

$$\frac{1 - 0.8475}{0.9524 + 0.9009 + 0.8475}$$

$$= 0.1525 / 2.7008 = 0.0565$$

(Module 30.6, LOS 30.e)

The Isle of Nefer is a developing country with its stock and futures markets enjoying record trading volumes due to the influx of foreign funds. You are looking to invest in the stock and futures markets in the Isle of Nefer. The representative stock market index, Nefer Industrial Index (NII), is currently priced at 8,765 and the one year NII future contract is currently trading at 8,920.

You have experience in using forward contracts but not futures. You discuss the possibility of investing in the Isle of Nefer using futures contract with your supervisor, Peter Filler, and he makes the following comments.

Comment 1: "A futures contract will have positive value after marking to market if the future price is up on that day."

Comment 2: "Given a quoted clean bond price of the CTD, when looking at a bond future the full price of the bond must be used which equals the clean price of the bond plus accrued interest. The futures price can then be calculated as:

$$QFP = \{(full\ price) \times (1 + R_f)^T + AI_T - FVC\}(1 / CF)$$

Where AI_T = accrued interest at futures maturity, R_f = risk-free rate, FVC = future value of coupon and CF = conversion factor

Peter Filler also suggests that you invest in Treasury bond futures. Exhibit 1 contains the relevant information.

Exhibit 1

Price of underlying deliverable 16 year 5% Treasury bond (just paid coupon) \$1,030

Expiration of Treasury bond futures contract

0.7 year

Conversion factor

1.08

Risk free rate

3.0 percent

Question #23 - 26 of 65

Question ID: 1473637

How will the price of the one-year stock index future perform over the next 12 months?

- A) The futures price will converge to the future spot index price, with the basis reducing to zero. 
- B) The value will move approximately in line with the spot index price, with a fairly constant basis. 
- C) The futures price will move approximately in line with the spot index price, though its actual level at the end of the year will depend more on supply and demand than on the spot price. 

Explanation

The futures price will converge to the spot price over the lifetime of the contract, with the basis (difference between future and spot) reducing over this period to zero as an expiring futures contract is the same as a spot transaction. At expiration the futures and spot price will converge to prevent an arbitrage opportunity. Answer B is wrong, as it implies the future price will move in a straight line.

(Module 30.3, LOS 30.d)

Question #24 - 26 of 65

Question ID: 1473638

Comment 1 is *best* described as:

- A) correct. 
- B) incorrect as the value of the futures contract should be negative after marking to market. 
- C) incorrect as the value of the futures contract should be zero after marking to market. 

Explanation

The value of a futures contract will reset to zero after marking-to-market.

(Module 30.3, LOS 30.d)

Question #25 - 26 of 65

Question ID: 1473639

Comment 2 is *best* described as:

- A) correct. 
- B) incorrect as the full price should be the clean price less the current accrued interest. 
- C) incorrect as the accrued interest at expiration should be deducted from the future value of full bond price in arriving at the quoted futures price. 

Explanation

The quoted (clean) price of the futures contract is the future value of the full price of the cheapest to deliver bond less accrued interest (since last coupon) and the future value any coupon payment received during the life of the contract and then adjusted for the appropriate conversion factor.

$$QFP = \{(\text{full price}) \times (1 + R_f)^T - AI_T - FVC\}(1 / CF)$$

(Module 30.3, LOS 30.d)

Question #26 - 26 of 65

Question ID: 1473640

Using Exhibit 1, the quoted price of the Treasury bond futures contract should be *closest* to:

- A) \$941. 
- B) \$975. 
- C) \$1,100. 

Explanation

An investor of the Treasury bond will receive one semi-annual coupon in 0.5 years from now (or 0.2 years before maturity). At expiration of the futures contract, the CTD bond will have 0.2 years of accrued interest.

$$\text{FV of coupon (FVC)} = \$25 \times 1.03^{(0.7 - 0.5)} = \$25.15$$

$$\text{AI}_T = 0.2 / 0.5 \times \$25 = \$10$$

$$\text{QFP} = \{(\text{full price}) \times (1 + R_f)^T - \text{AI}_T - \text{FVC}\}(1 / \text{CF})$$

$$= [\$1030(1.03)^{0.7} - 10 - 25.15](1 / 1.08)$$

$$= \$941.10$$

(Module 30.3, LOS 30.d)

Question #27 of 65

Question ID: 1473615

The price of a forward contract:

- A)** changes over the term of the contract.
- B)** depends on forward interest rates.
- C)** is determined at contract initiation.



Explanation

The price of a forward contract is established at the initiation of the contract and is expressed in different terms, depending on the underlying assets. It is the price that makes the contract value zero, and depends on current interest rates through the cost-of-carry calculation.

(Module 30.1, LOS 30.b)

Question #28 of 65

Question ID: 1473654

At the inception of a market-rate plain vanilla swap, the value of the swap to the fixed-rate payer is:

- A)** positive.
- B)** zero.
- C)** either positive or negative.



Explanation

A market-rate swap is priced so that the value to either side is zero at the inception of the swap.

(Module 30.6, LOS 30.e)

Question #29 of 65

Question ID: 1473652

The price and value of a plain vanilla interest-rate swap are:

- A) never equal. 
- B) equal in equilibrium. 
- C) only equal at the inception of a swap contract. 

Explanation

The price of a swap is the fixed rate specified in the swap and is the same for the payer and the receiver. The value is the dollar value of the contract to the fixed-rate payer and is the opposite of the value to the floating-rate payer.

(Module 30.6, LOS 30.e)

Question #30 of 65

Question ID: 1473668

A U.S. firm (U.S.) and a foreign firm (F) engage in a 3-year, annual pay currency swap; The USD fixed rate at initiation was 5% while FC fixed rate was 4%. At the beginning of the swap, \$2 million was paid by the U.S. firm and the exchange rate was 2 FC units per \$1. At the end of the swap period the exchange rate was 1.75 FC units per \$1.

At the end of year 1, firm:

- A) U.S. pays firm F \$200,000. 
- B) F pays firm U.S. \$200,000. 
- C) U.S. pays firm F 160,000 FC units. 

Explanation

Firm U.S. pays fixed 4% on FC = $0.04 \times \$2,000,000 \times 2$ FC units per \$1 = 160,000 FC units.

(Module 30.7, LOS 30.f)

Question #31 of 65

Question ID: 1473642

The price of a 3×5 forward rate agreement (FRA) is the:

- A) 2-month implied forward rate 3 months from today.
- B) 2-month implied forward rate 5 months from today.
- C) 3-month implied forward rate 5 months from today.



Explanation

The notation for FRAs is unique. There are two numbers associated with an FRA: the number of months until the contract expires and the number of months until the underlying loan is settled. The difference between these two is the maturity of the underlying loan. For example, a 3×5 FRA is a contract that expires in three months (90 days), and the underlying loan is settled in five months (150 days). The price of the 3×5 FRA is calculated by annualizing the implied forward rate. The implied forward rate is calculated from the 3-month rate and the 5-month rate.

(Module 30.4, LOS 30.c)

Question #32 of 65

Question ID: 1473684

Consider a 1-year semiannual equity swap based on an index at 985 and a fixed rate of 4.4%. 90 days after the initiation of the swap, the index is at 982 and MRR is 4.6% for 90 days and 4.8% for 270 days. The value of the swap to the equity payer, based on a \$2 million notional value is *closest* to:

- A) -\$22,564
- B) \$22,314
- C) \$22,564



Explanation

$$\begin{aligned}
&= \frac{982}{985} - \frac{\frac{0.044}{2}}{1 + \left(0.046 \times \frac{90}{360}\right)} - \frac{\frac{0.044}{2}}{1 + \left(0.048 \times \frac{270}{360}\right)} - \frac{1}{1 + \left(0.048 \times \frac{270}{360}\right)} \\
&= \frac{982}{985} - \frac{0.022}{1.0115} - \frac{0.022}{1.036} - \frac{1}{1.036} \\
&= 0.996954 - 0.021750 - 0.021236 - 0.965251 \\
&= -0.0112821 \times 2,000,000 = -\$22,564
\end{aligned}$$

-\$22,564 is the value to the fixed-rate payer, thus \$22,564 is the value to the equity return payer.

(Module 30.8, LOS 30.g)

Question #33 of 65

Question ID: 1473624

The value of a futures contract between the times when the account is marked-to-market is:

- A)** never less than the value of a forward contract entered into on the same date. ✗
- B)** equal to the difference between the price of a newly issued contract and the settle price at the most recent mark-to-market period. ✓
- C)** the same as the contract price. ✗

Explanation

Between the mark-to-market account adjustments, the contract value is calculated just like that of a forward contract; it is the difference between the price at the last mark-to-market and the current futures price, (i.e. the futures price on a newly issued contract). The mark-to-market of a futures contract is the payment or receipt of funds necessary to adjust for the gains or losses on the position. This adjusts the contract price to the 'no-arbitrage' price currently prevailing in the market.

(Module 30.1, LOS 30.b)

Question #34 of 65

Question ID: 1473620

The theoretical price of a forward contract:

- A)** is the no-arbitrage price. ✓
- B)** equals the long's expectation of the future price of the underlying asset. ✗

C) is always greater than the current price of the underlying asset.



Explanation

The theoretical price of a forward contract is the future price of the underlying asset imposed by the no-arbitrage conditions. It can be less than the current price of the asset if the cost-of-carry is negative. Accrued interest is paid by the long at delivery under a bond forward, but is not included in the price quote, which is usually in terms of yield to maturity at the settlement date.

(Module 30.1, LOS 30.b)

Question #35 of 65

Question ID: 1473616

At expiration, the value of a forward contract is:

A) equal to the market price of the underlying asset.



B) always greater than or equal to zero.



C) the difference between the contract price and the market value of the underlying asset.



Explanation

In a forward contract, the long is obligated to buy, and the short is obligated to sell, the underlying asset at the contract price. The difference between the contract price and the market price of the asset is what gives the contract value. The contract has a positive value at expiration to the long/short only if the contract price is below/above the market price.

(Module 30.1, LOS 30.b)

Question #36 of 65

Question ID: 1473673

90 days ago the exchange rate was USD 0.83 per CDN and the term structure was:

	180 days	360 days
USD MRR	5.2%	5.6%
CDN MRR	4.8%	5.4%.

A 1 year, semi-annual settlement, fixed for fixed swap was initiated with 5.30% fixed for CDN and 5.52% fixed for USD on a principal of USD 1 million.

Current exchange rate is USD 0.84 per CDN and the yield curve is:

	90 days	270 days
USD MRR	5.2%	5.6%
Disc Factor	0.98717	0.95969
CDN MRR	4.8%	5.4%
Disc factor	0.98814	0.96108

What is the value of the swap to the USD interest payer?

A) \$11,500.



B) -\$3,472.



C) \$10,126.



Explanation

CDN principal = $1,000,000 / 0.83 = \text{CDN } 1,204,819$. CDN payments = $(0.053/2) \times 1,204,819 = \text{CDN } 31,928$.

USD payment = $1,000,000 \times (0.0552/2) = \text{USD } 27,600$

Value of CDN payments = $31,928 \times 0.98814 + (1,204,819 + 31,928) \times 0.96108 = \text{CDN } 1220162$

Value in USD = $1220162 \times 0.84 = \text{USD } 1,024,936$.

Value of USD payments = $27,600 \times 0.98717 + 1,027,600 \times 0.95969 = \text{USD } 1,013,423$

Value to USD interest payer = $\$1,024,936 - \$1,013,423 = \$11,513$

(Module 30.7, LOS 30.f)

An equity portfolio manager who has a positive long-term outlook for equities, but expects equity prices to decline over the next three months, would *most appropriately* enter into:

- A) an equity swap as the equity return payer. 
- B) an equity swap as the equity return receiver. 
- C) a short straddle with an equity index as the underlying. 

Explanation

By entering an equity swap as the equity return payer, the manager can protect the portfolio value from a short-term decline in equity prices while keeping ownership of the equities for the longer term. A short straddle costs the investor when the price of the underlying moves up or down.

(Module 30.8, LOS 30.g)

Question #38 of 65

Question ID: 1473627

To initiate an arbitrage trade if the futures contract is underpriced, the trader should:

- A) short the asset, invest at the risk-free rate, and buy the futures. 
- B) borrow at the risk-free rate, buy the asset, and sell the futures. 
- C) borrow at the risk-free rate, short the asset, and sell the futures. 

Explanation

If the futures price is too low relative to the no-arbitrage price, buy futures, short the asset, and invest the proceeds at the risk-free rate until contract expiration. Take delivery of the asset at the futures price, pay for it with the loan proceeds and keep the profit. For Treasury bill (T-bills), shorting the asset is equivalent to borrowing at the T-bill rate.

(Module 30.1, LOS 30.b)

Question #39 of 65

Question ID: 1473662

Writing a series of interest-rate puts and buying a series of interest-rate calls, all at the same exercise rate, is equivalent to:

- A) being the floating-rate payer in an interest rate swap. 
- B) being the fixed-rate payer in an interest rate swap. 
- C) a short position in a series of forward rate agreements. 

Explanation

A short position in interest rate puts will have a negative payoff when rates are below the exercise rate; the calls will have positive payoffs when rates exceed the exercise rate. This mirrors the payoffs of the fixed-rate payer who will receive positive net payments when settlement rates are above the fixed rate.

(Module 30.6, LOS 30.e)

Question #40 of 65

Question ID: 1473669

The current U.S. dollar (\$) to Canadian dollar (C\$) exchange rate is 0.7. In a \$1 million currency swap, the party that is entering the swap to hedge existing exposure to C\$-denominated fixed-rate liability will:

- A) pay floating in C\$. 
- B) pay C\$1,428,571 at the beginning of the swap. 
- C) receive floating in C\$. 

Explanation

The party that is entering the swap to hedge existing exposure to C\$-denominated fixed-rate liability will want to receive-fixed C\$. They will pay $1,000,000/0.7 = \text{C\$}1,428,571$ (principal) at swap inception (in exchange for USD 1 million) and get the same amount (C\$1,428,571) back at termination (in exchange for paying back the USD 1 million).

(Module 30.7, LOS 30.f)

Question #41 of 65

Question ID: 1473621

At contract initiation, the value of a forward contract:

- A) is typically zero regardless of the price of the underlying asset. 
- B) depends on the market price of the underlying asset. 
- C) is set to 100 by convention. 

Explanation

Due to the no-arbitrage principle, the price of a forward contract is calculated to make the value of the contract zero at contract initiation. Neither the long nor the short typically makes any payment to enter into the forward agreement. A special case is an off-market forward where, for whatever reason, the contract price is not set equal to the no-arbitrage price, and the long or short position makes a payment to the opposite counterparty to offset the difference.

(Module 30.1, LOS 30.b)

Question #42 of 65

Question ID: 1473658

For a 1-year quarterly-pay swap, an equivalent position with short puts and long calls would involve:

- A) three put-call combinations on the last three settlement dates of the swap. 
- B) three put-call combinations expiring on the first three settlement dates of the swap. 
- C) put-call combinations expiring on each of the four settlement dates. 

Explanation

Interest rate options pay one period after exercise. Options expiring on settlements at $t = 1, 2, 3$, will mimic the uncertain swap payments at $t = 2, 3, 4$.

(Module 30.6, LOS 30.e)

Elodie Brodeur works in the finance department of a large fashion house in France. The international catwalk season will start in two months' time and Brodeur has worked out that the company will need a 3-month loan of €4m in two months' time. The company's lenders are typically retail banks offering loans at MRR plus 40 basis points. Brodeur is concerned that interest rates may rise during the next two months and wants to use a FRA to lock-in the borrowing cost. Brodeur has collected the rate information shown in Exhibit 1.

Exhibit 1: Current MRR Rates

60 day MRR 2.0%

90 day MRR 2.4%

150 day MRR 2.6%

210 day MRR 2.9%

One month after the initiation of the FRA the MRR rates are shown in Exhibit 2.

Exhibit 2: MRR Rates One Month Later

30 day MRR 1.8%

90 day MRR 2.5%

120 day MRR 2.8%

180 day MRR 3.0%

At the expiration of the FRA, 90-day MRR is 3.4%.

Question #43 - 46 of 65

Question ID: 1473645

Which of the following comments relating to Brodeur's use of a forward rate agreement is *least* accurate?

- A) A short FRA can be used to lock into a fixed rate of borrowing commencing in two months' time and expiring in five months' time. 
- B) The use of a FRA to hedge interest rate risk would lock Brodeur into paying a fixed rate plus 40 basis points for her borrowing. 
- C) The use of a FRA to hedge interest rate risk on her future loan will mean that she no longer benefits if interest rates fall. 

Explanation

A long FRA will create the obligation to pay fixed and receive floating from months 2 to month 5. The floating payments will offset her bank loan leaving her with a pay fixed obligation, where the fixed rate is determined today.

The floating receipt on the FRA and the floating payment on the loan she will need to take out will offset leaving a net payment of 40 basis points. Overall, she will now pay the fixed rate on the swap plus the 40 basis points on the loan.

The use of a long FRA will lock Elodie into paying fixed interest rate on the FRA (the FRAs forward price) plus the basis points on her loan above MRR. Elodie will no longer benefit from interest rate declines but will be protected from interest rises.

(Module 30.4, LOS 30.c)

Question #44 - 46 of 65

Question ID: 1473646

Using the data in Exhibit 1, which of the following is *closest* to the forward price of the FRA?

A) 2.4%.



B) 2.8%.



C) 3.0%.



Explanation

Step 1: Identify the correct MRR rates.

We will require the MRR rate until FRA expiry (day 60) and also the MRR rate at the end of the borrowing/lending period (day 150).

The 90-day MRR rate is a distractor. Elodie will borrow for 90 days but not from current date (T0). Elodie requires a 90-day loan commencing in 60 days' time.

Step 2: Unannualize the quoted rates.

$$\text{MRR}_{60 \text{ day}} = 2\% \times \frac{60}{360} = 0.3333\%$$

$$\text{MRR}_{150 \text{ day}} = 2.6\% \times \frac{150}{360} = 1.0833\%$$

Step 3: Compute the annualized forward price (fixed rate) starting in 60 days and lasting for 90 days.

$$\text{Forward rate} = ((1 + \text{long rate}) / (1 + \text{short rate}) - 1)(360 / 90) = 2.99\%$$

(Module 30.4, LOS 30.c)

For this question only, assume that the forward price of the FRA was 2.9%. Which of the following is the closest to the value accrued on the FRA from Brodeur's perspective one month after initiation of the contract?

A) +€2,300.



B) +€2,200.



C) -€2,265.



Explanation

The value of a FRA before expiry can be calculated by comparing the fixed rate on the original FRA to the fixed rate on a new FRA covering the same borrowing and lending period.

Step 1: Identify the correct MRR rates.

After 1 month has passed there is now 1 month until the original FRAs expiry and 4 months to the end of the borrowing and lending period. So we will require 30 day and 120 day MRR (i.e., (1×4) FRA price)

Step 2: Unannualize rates.

$$\text{MRR}_{30 \text{ day}} = 1.8\% \times \frac{30}{360} = 0.15\%$$

$$\text{MRR}_{120 \text{ day}} = 2.8\% \times \frac{120}{360} = 0.9333\%$$

Step 3: Compute the fixed rate (forward price) on a new FRA with the same expiry as the original FRA.

$$\text{Forward rate} = ((1 + \text{long rate}) / (1 + \text{short rate}) - 1)(360 / 90) = 3.1286\%$$

Step 4: Compute the gain/(loss) on FRA at the end of the borrowing/lending period.

$$\text{Gain to long} = (\text{new fixed rate} - \text{original fixed rate})(\text{days} / 360)(\text{notional})$$

$$= (0.0313 - 0.029)(90 / 360)(4\text{m}) = 2,300$$

Step 5: Discount the gain/(loss) from the end of borrowing and lending to the valuation date (120 days).

$$\frac{€2,300}{1.009333} = €2,278$$

(Module 30.4, LOS 30.c)

Assuming the forward price of the FRA was 2.9%, which of the following is *closest* to the value of the FRA at expiration?

A) €4,958.



B) €5,000.



C) €19,831.



Explanation

Step 1: Compute gain/(loss) at end of borrowing and lending period.

$$(\text{floating rate} - \text{fixed rate}) \left(\frac{90}{360} \right) (\text{notional principal})$$

$$(0.034 - 0.029) \left(\frac{90}{360} \right) (\text{€4m}) = \text{€5,000}$$

Step 2: Discount the gain/(loss) back to FRA expiry (90 days).

$$\text{MRR}_{90 \text{ day}} = 3.4\% \times \frac{90}{360} = 0.85 \%$$

$$\frac{\text{€5,000}}{1.0085} = \text{€4,958}$$

(Module 30.4, LOS 30.c)

Question #47 of 65

Question ID: 1473657

A swap is equivalent to a series of:

A) FRAs priced at market rates.



B) off-market FRAs.



C) interest rate calls.



Explanation

Since the fixed rate on the swap is the same at every settlement date, a series of FRAs at those fixed rates will have values that differ from zero to the extent the fixed rate and the zero-value rate differ. This makes them off-market FRAs.

(Module 30.6, LOS 30.e)

Question #48 of 65

Question ID: 1473630

An index is currently 965 and the continuously compounded dividend yield on the index is 2.3%. What is the no-arbitrage price on a one-year index forward contract if the continuously compounded risk-free rate is 5%.

A) 991.4.



B) 991.1.



C) 987.2.



Explanation

The futures price $FP = S_0 e^{-\delta T} (e^{RT})$

$$= S_0 e^{(R-\delta)T}$$

$$= 965 e^{(.05-.023)}$$

$$= 991.4$$

(Module 30.2, LOS 30.a)

Question #49 of 65

Question ID: 1473626

During the life of a forward contract, the value of the contract is *best* described as:

A) the difference between the spot price and the present value of the forward price of the underlying asset.



B) the difference between the future value of the spot price and the expected future price of the underlying asset.



C) the present value of the expected future price of the underlying asset.



Explanation

The value of a forward contract on an asset with no cash flows during its term is equal to spot – (forward price) / $(1 + R_f)^t$, the difference between the spot price and the present value of the forward price of the underlying asset.

(Module 30.1, LOS 30.b)

Question #50 of 65

Question ID: 1473671

Consider a fixed-for-fixed 1-year \$100,000 semiannual currency swap with rates of 5.0% in USD and 4.8% in CHF, originated when the exchange rate is \$0.34. After the first settlement, the exchange rate is \$0.35 and the term structure is:

	90 days	180 days
MRR	5.2%	5.6%
Swiss	4.8%	5.4%

What is the value of the swap to the USD payer?

A) \$2,937



B) -\$2,719.



C) \$2,814.



Explanation

CHF periodic coupon (per 1 CHF) = $0.048/2 = 0.024$

DF for 180 day CHF = $1 / (1 + 0.054 \times (180/360)) = 1/1.027 = 0.9737$

PV of CHF cash flows (per 1 CHF) = $0.9737 \times 1.024 = 0.9971$

At the current exchange rate the value is $0.9971 \times 0.35 = \text{USD } 0.3490$

The notional amount is $100,000/0.34 = 294,118$ CHF so the dollar value of the CHF payments is $0.3490 \times 294,118 = \$102,647$.

USD periodic coupon (per 1 USD) = $0.05/2 = 0.025$

DF for 180 day USD = $1 / (1 + 0.056 \times (180/360)) = 1/1.028 = 0.9728$

PV of USD cash flows (per 1 USD) = $0.9728 \times 1.025 = 0.9971$

Value (for notional = \$100,000) = $0.9971 \times 100,000 = \$99,710$.

The value of the swap to the dollar payer is $102,647 - \$99,710 = \$2,937$.

(Module 30.7, LOS 30.f)

Question #51 of 65

Question ID: 1473674

In an equity return swap for MRR, if the return on the underlying equity portfolio is negative for a payment period, the equity return payer will:

A) receive a net payment less than the loss of value on the equity portfolio.



B) receive a net payment greater than the loss of value on the equity portfolio.



C) make a net payment greater than the loss of value on the equity portfolio.



Explanation

The equity return payer will receive the periodic interest payment and "pay" the negative return on the portfolio, resulting in a net payment to the equity return payer that is greater than the loss on the equity portfolio.

(Module 30.8, LOS 30.g)

Question #52 of 65

Question ID: 1473663

Which of the following is *equivalent* to a pay-fixed swap with a tenor of two years with semi-annual swap payments and a fixed rate of 6% (exchanged for MRR)? The notional principal is \$100,000,000.

- A strip of three forward rate agreements, which obligates the party to pay a
A) fixed rate of 6% and receive six-month MRR on a notional principal of 
\$100,000,000.
- B)** A strip of two forward rate agreements, which obligates the party to pay a fixed
rate of 6% and receive six-month MRR on a notional principal of \$100,000,000. 
- C)** A forward rate agreement, which obligates the party to pay a fixed rate of 6%
and receive six-month MRR on a notional principal of \$100,000,000. 

Explanation

In an interest rate swap, the first payment is known with certainty and will be made at month 6. The determination dates for the floating rate will be at months 6, 12, and 18 and the corresponding payment dates will be at months 12, 18, and 24. These correspond to the three forward rate agreements.

(Module 30.6, LOS 30.e)

Question #53 of 65

Question ID: 1473632

A stock is currently priced at \$110 and will pay a \$2 dividend in 85 days and is expected to pay a \$2.20 dividend in 176 days. The no arbitrage price of a six-month (182-day) forward contract when the effective annual interest rate is 8% is *closest* to:

- A)** \$110.00. 
- B)** \$110.06. 
- C)** \$110.20. 

Explanation

In the formulation below, the present value of the dividends is subtracted from the spot price, and then the future value of this amount at the expiration date is calculated.

$$(110 - 2/1.08^{85/365} - 2.20/1.08^{176/365}) 1.08^{182/365} = \$110.06$$

Alternatively, the future value of the dividends could be subtracted from the future value of the stock price based on the risk-free rate over the contract term.

(Module 30.2, LOS 30.a)

Question #54 of 65

Question ID: 1473667

Which of the following is equivalent to a pay-floating USD receive-fixed EUR currency swap position?

- A) A short position in a EUR bond coupled with the issuance of a USD-denominated floating rate note. 
- B) A short position in a EUR bond coupled with a long position in a USD-denominated floating rate note. 
- C) A long position in a EUR bond coupled with the issuance of a USD-denominated floating rate note. 

Explanation

A long position in a fixed rate EUR bond will receive fixed coupons denominated in EUR. The short floating rate note requires USD denominated floating-rate payments. Combined, these are the same cash flow as a pay-floating USD receive-fixed EUR currency swap.

(Module 30.7, LOS 30.f)

Question #55 of 65

Question ID: 1473631

Calculate the no-arbitrage forward price for a 90-day forward on a stock that is currently priced at \$50.00 and is expected to pay a dividend of \$0.50 in 30 days and a \$0.60 in 75 days. The annual risk free rate is 5% and the yield curve is flat.

- A) \$48.51. 
- B) \$50.31. 
- C) \$49.49. 

Explanation

The present value of expected dividends is: $\$0.50 / (1.05^{30 / 365}) + \$0.60 / (1.05^{75 / 365}) = \1.092

Future price = $(\$50.00 - 1.092) \times 1.05^{90 / 365} = \49.49

(Module 30.2, LOS 30.a)

Question #56 of 65

Question ID: 1473681

Consider a fixed-rate semiannual-pay equity swap where the equity payments are the total return on a \$1 million portfolio and the following information:

- 180-day MRR is 5.2%
- 360-day MRR is 5.5%
- Dividend yield on the portfolio = 1.2%

What is the fixed rate on the swap?

A) 5.4234%.



B) 5.4197%.



C) 5.1387%.



Explanation

$$\frac{\left(1 - \frac{1}{1.055}\right)}{\left(\frac{1}{1 + 0.052\left(\frac{180}{360}\right)} + \frac{1}{1 + 0.055\left(\frac{360}{360}\right)}\right)} = 0.027117 \times 2 = 5.4234\%$$

(Module 30.8, LOS 30.g)

Question #57 of 65

Question ID: 1473650

30 days ago, J. Klein took a short position in a \$10 million (3X6) forward rate agreement (FRA) based on MRR and priced at 5%. The current MRR curve is:

- 30-day = 4.8%
- 60-day = 5.0%
- 90-day = 5.1%
- 120-day = 5.2%
- 150-day = 5.4%

The current value of the FRA, to the short, is *closest* to:

A) -\$15,280.



B) -\$15,495.



C) -\$15,154.



Explanation

FRAs are entered in to hedge against interest rate risk. A person would buy a FRA anticipating an increase in interest rates. If interest rates increase more than the rate agreed upon in the FRA (5% in this case) then the long position is owed a payment from the short position.

Step 1: Find the forward 90-day MRR 60-days from now.

$[(1 + 0.054(150 / 360)) / (1 + 0.05(60 / 360)) - 1](360 / 90) = 0.056198$. Since projected interest rates at the end of the FRA have increased to approximately 5.6%, which is above the contracted rate of 5%, the short position currently owes the long position.

Step 2: Find the interest differential between a loan at the projected forward rate and a loan at the forward contract rate.

$$(0.056198 - 0.05) \times (90 / 360) = 0.0015495 \times 10,000,000 = \$15,495$$

Step 3: Find the present value of this amount 'payable' 90 days after contract expiration (or 60 + 90 = 150 days from now) and note once again that the short (who must 'deliver' the loan at the forward contract rate) loses because the forward 90-day MRR of 5.6198% is greater than the contract rate of 5%.

$$[15,495 / (1 + 0.054(150 / 360))] = \$15,154.03$$

This is the *negative* value to the short.

(Module 30.5, LOS 30.c)

Question #58 of 65

Question ID: 1473618

Which of the following *best* describes the price of a forward contract? The forward price is:

A) always equal to the market price at contract termination.



B) the price that makes the values of the long and short positions zero at contract initiation. 

C) always expressed in dollars. 

Explanation

The forward price is the contract price of the underlying asset under the terms of the forward contract, and is the price that makes the values of the long and short positions zero at contract initiation. It is not the amount it costs to purchase the forward contract. The forward price is expressed in terms of the underlying asset, and may be a dollar value, exchange rate, or interest rate. The value of a forward contract comes from the difference between the forward contract price and the market price for the underlying asset. These values are likely to be different at contract termination, which will result in a profit for either the long or the short position.

(Module 30.1, LOS 30.b)

Question #59 of 65

Question ID: 1473655

The fixed-rate on a semiannual 2-year interest rate swap is *closest* to the:

A) current 180-day T-bill rate. 

B) coupon rate on a 2-year par bond with the same credit risk as the reference rate. 

C) coupon rate on a 2-year par bond with the same credit risk as the fixed-rate payer. 

Explanation

The fixed-rate on a swap is calculated using the yield curve for the floating rate reference. Therefore, the fixed rate reflects the credit spread of that rate over the riskless rate of return.

(Module 30.6, LOS 30.e)

Question #60 of 65

Question ID: 1473651

The floating-rate payer in a simple interest-rate swap has a position that is equivalent to:

A) a series of long forward rate agreements (FRAs). 

B) issuing a floating-rate bond and a series of long FRAs. 

C) a series of short FRAs. 

Explanation

The floating-rate payer has a liability/gain when rates increase/decrease above the fixed contract rate; the short position in an FRA has a liability/gain when rates increase/decrease above the contract rate.

(Module 30.5, LOS 30.c)

Question #61 of 65

Question ID: 1473629

The value of the S&P 500 Index is 1,260. The continuously compounded risk-free rate is 5.4% and the continuous dividend yield is 3.5%. Calculate the no-arbitrage price of a 160-day forward contract on the index.

A) \$1,310.13.



B) \$1,270.54.



C) \$562.91.



Explanation

$$FP = 1,260 \times e^{(0.054 - 0.035) \times (160 / 365)} = 1,270.54$$

(Module 30.2, LOS 30.a)

Question #62 of 65

Question ID: 1473682

Consider a fixed-rate semiannual-pay equity swap where the equity payments are the total return on a \$1 million portfolio and the following information:

- 180-day MRR is 4.2%
- 360-day MRR is 4.5%
- Div. yield on the portfolio = 1.2%

What is the fixed rate on the swap?

A) 4.4477%.



B) 4.5143%.



C) 4.3232%.



Explanation

$$\frac{\left(1 - \frac{1}{1.045}\right)}{\left(\frac{1}{1 + 0.042\left(\frac{180}{360}\right)} + \frac{1}{1 + 0.045\left(\frac{360}{360}\right)}\right)} = 0.022239 \times 2 = 4.4477 \%$$

(Module 30.8, LOS 30.g)

Question #63 of 65

Question ID: 1473661

The fixed-rate payer in an interest-rate swap has a position equivalent to a series of:

- A) short interest-rate puts and long interest-rate calls. ✓
- B) long interest-rate puts and calls. ✗
- C) long interest-puts and short interest-rate calls. ✗

Explanation

The fixed-rate payer has profits when short rates rise and losses when short rates fall, equivalent to writing puts and buying calls.

(Module 30.6, LOS 30.e)

Question #64 of 65

Question ID: 1473623

What is the difference between spot and futures prices? Spot prices are always:

- A) equal to the futures price at futures expiration. ✓
- B) delivered to meet the futures obligation at expiration. ✗
- C) lower than futures prices. ✗

Explanation

The difference between the spot and the futures price must be zero at expiration to avoid arbitrage.

(Module 30.1, LOS 30.b)

Question #65 of 65

Question ID: 1473656

A plain vanilla interest-rate swap to the fixed-rate payer is equivalent to issuing a fixed-rate bond and:

A) selling a series of interest rate puts.



B) selling a series of interest rate calls.



C) buying a floating-rate bond.



Explanation

Paying fixed and receiving floating in a swap is equivalent to issuing a fixed-rate bond and investing the proceeds in a floating rate bond.

(Module 30.6, LOS 30.e)